EVALUATION REPORT: ROUTER AUTO-SELECTION VS HEAVY-ONLY LLM

Generated: 2026-10-20T09:59:14.311Z | Run ID: 2026-10-20T09-58-47-702Z | Tests: 30

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1. EXECUTIVE SUMMARY & EFFICIENCY VERDICT

Verdict: Substantial Speedup: Router yielded 25.1% latency advantage over heavy-only execution.

Efficiency Score: 44.4 / 100

Heavy Baseline Model: llama3.1:8b

2. LATENCY & THROUGHPUT COMPARISON

â¢ Router Mean Latency: 24.7 s (p50: 13.3 s)

â¢ Heavy-Only Mean Latency: 33.0 s (p50: 30.5 s)

â¢ Latency Reduction: 25.1% efficiency gain

â¢ Average Tokens/Sec: Router 72.0 tok/s vs Heavy 16.8 tok/s

3. TOKEN USAGE & DIFFERENCE

â¢ Router Tokens: 14990 total (1208 prompt + 13782 completion)

â¢ Heavy Tokens: 15104 total (790 prompt + 14314 completion)

â¢ Token Usage Difference: Router used 114 fewer tokens (0.8% difference)

4. ROUTER TIER DISTRIBUTION

â¢ Light Tier: 13 (43%)

â¢ Code Tier: 9 (30%)

â¢ Heavy Tier: 8 (27%)

5. DETAILED TEST CASE BREAKDOWN (Top Cases)

Test ID	Router Tier	R-Lat(ms)	H-Lat(ms)	Saved(ms)	Eff %	Tok Diff
eval-001	light	3210	7616	4405	58%	7
eval-002	light	3839	35181	31341	89%	180
eval-003	light	2470	6119	3649	60%	-3
eval-004	light	2540	5067	2526	50%	-19
eval-005	light	2180	5844	3664	63%	-10
eval-006	light	2525	5269	2743	52%	-19
eval-007	light	2918	16126	13208	82%	77
eval-008	light	2138	4243	2105	50%	-19
eval-009	light	2149	4668	2519	54%	-19
eval-010	light	2156	6246	4090	65%	13
eval-011	code	26801	23441	-3360	-14%	10
eval-012	code	30894	32831	1937	6%	-93
eval-013	code	8049	13247	5198	39%	46
eval-014	code	13253	26450	13196	50%	170
eval-015	code	28845	30505	1660	5%	-94
eval-016	code	13080	43633	30553	70%	468

eval-017	light	7987	31393	23406	75%	-137
eval-018	code	23858	30434	6576	22%	12
eval-019	code	30229	30543	314	1%	-123
eval-020	code	16976	29786	12810	43%	125
eval-021	heavy	63671	82611	18941	23%	289
eval-022	heavy	51650	46586	-5064	-11%	-76
eval-023	heavy	59058	55854	-3204	-6%	-41
eval-024	heavy	96552	71384	-25168	-35%	-338
eval-025	heavy	26871	20382	-6489	-32%	-123
eval-026	light	7775	54009	46234	86%	41
eval-027	heavy	82417	68651	-13766	-20%	-200
eval-028	heavy	63928	76461	12532	16%	193
eval-029	heavy	53946	59725	5779	10%	62
eval-030	light	10260	66452	56192	85%	-265